

Solving SAT in polynomial time?

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March 22, 2025

Abstract

Our goal is to shed some light on the $\mathcal{P}$ vs $\mathcal{NP}$ problem by looking at a SAT solver and investigate whether it runs in polynomial time. Given the $\mathcal{NP}$ -completeness of SAT, if it does, this would imply $\mathcal{P} = \mathcal{NP}$.

Keywords: $\mathcal{P}$ vs $\mathcal{NP}$, polynomial time SAT solver, Boolean algebra

We would like to acknowledge all of the many people who contributed to this work.

1 A SAT solver which might run in polynomial time

1.1 Input Boolean wff φ

Either $\varphi = p_i$ for a propositional variable p_i or $\varphi = \neg\varphi_1$ for some Boolean wff φ_1 , or $\varphi = \varphi_1 \wedge \varphi_2$ for Boolean wffs φ_1 and φ_2 , or $\varphi = \varphi_1 \vee \varphi_2$ for Boolean wffs φ_1 and φ_2 .

1.2 Compute a polynomial associated with φ called $\tau(\varphi)$ as follows:

$$\tau(p_i) := x_i,$$

$$\tau(\neg\varphi_1) := 1 + \tau(\varphi_1),$$

$$\tau(\varphi_1 \wedge \varphi_2) := \tau(\varphi_1)\tau(\varphi_2),$$

$$\tau(\varphi_1 \vee \varphi_2) := 1 + (1 + \tau(\varphi_1))(1 + \tau(\varphi_2)).$$

Calculations are done in $\mathbb{F}_2$.

1.3 Let $J = \langle \tau + 1, x_1^2 + x_1, \dots, x_n^2 + x_n \rangle$ be an ideal in $\mathbb{F}_2[x_1, \dots, x_n]$

Note that

$$J = \langle \tau + 1 \rangle + \langle x_1^2 - x_1, \dots, x_n^2 - x_n \rangle.$$

- 1.4 Find a Gröbner basis G for J which might be done in polynomial time using the FGLM algorithm or a different algorithm
- 1.5 Check whether $\bar{1}^G = 0$
 - 1.5.1 If $\bar{1}^G = 0$, then φ is not satisfiable
 - 1.5.2 If $\bar{1}^G \neq 0$, then φ is satisfiable